LAIBA FAROOQ

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PEC No: BIOMEDICAL/003296

OBJECTIVE

Biomedical Engineer with a Master's degree in progress, a peer-reviewed publication and hands-on industry experience in medical device installation, calibration and maintenance. Seeking a Lecturer position in Biomedical Engineering to teach and mentor students in medical instrumentation, biomedical signal processing, biomaterials and biomechanics, and to contribute to research in affordable healthcare technology.

EDUCATION

Master of Science in Biomedical Engineering *In progress*

NED University of Engineering and Technology, Karachi, PK

Bachelor of Engineering in Biomedical Engineering *August 2024*

NED University of Engineering and Technology, Karachi, PK | Cumulative GPA: 3.682

Intermediate in Pre-Engineering *April 2020*

Khatoon-e-Pakistan Govt. Degree College for Women, Karachi, PK | Grade: A

Matriculation in Science *March 2018*

Azmi Model School Secondary, Karachi, PK | Grade: A-1

RESEARCH AND PUBLICATION

"Affordable Design and Implementation of a 4-Channel EEG Bio-signal Amplification System with Mobile App Visualization Interface", *AMRJ Journal*, published 30 June 2024.

Final Year Project: 4-Channel EEG Bio-signal Amplification System with Mobile App Interface

- Developed a cost-effective, user-friendly EEG system for brain disease monitoring.
- Built a 4-channel EEG amplification system with a mobile app for real-time data visualization and analysis.
- Reduced the cost barrier for EEG-based monitoring and expanded accessibility for healthcare providers and patients.
- Enhanced brain disease diagnosis and monitoring capabilities.

TEACHING, TRAINING AND MENTORING

- Trained ophthalmologists and technicians on the correct operation, cleaning and routine care of ophthalmic equipment.
- Presented research as a poster presenter at the 3rd International Biomedical and Digital Health Conference 2024.
- Volunteered at the 1st International Biomedical and Digital Health Conference 2022.
- Prepared technical service reports and documentation, communicating complex technical information to clinical and non-technical audiences.

PROFESSIONAL EXPERIENCE

Service Engineer and Coordinator *16 Dec 2024 – present*

Jasani Scientifics, Karachi, PK

- Installed, calibrated and maintained ophthalmic units including slit lamps, lasers, autorefractometers, phaco units and other diagnostic devices at clinics and hospitals.
- Identified and resolved technical faults in ophthalmic instruments to ensure accurate performance and minimal downtime.
- Coordinated service schedules with eye clinics, managed repair timelines, and liaised between clients and technical teams.
- Trained ophthalmologists and technicians on proper operation and routine care of equipment.
- Prepared service reports, maintained equipment history records, and tracked warranty and spare part usage.

Biomedical Engineering Intern *1 – 30 Sep 2023*

Patel Hospital, Karachi, PK

- Gained hands-on experience with medical devices and electronics while collaborating with a dedicated team in a dynamic healthcare environment.

Biomedical Engineering Intern *27 Mar – 21 Apr 2023*

Dow Hospital (Ojha Campus), Karachi, PK

- Gained hands-on experience in Critical Care, Laboratory, Dental and Radiology departments.
- Operated, repaired and maintained various medical devices, ensuring optimal functionality in fast-paced clinical environments.

Biomedical Engineering Intern *12 Sep – 10 Oct 2022*

Abbasi Shaheed Hospital, Karachi, PK

- Visited Critical Care, Laboratory, Dental and Radiology departments and studied the technical specifications, applications and operating procedures of various medical machines.

ACADEMIC PROJECTS

- **Distillation Plant:** Designed automated switching between distillation stages and a real-time temperature display system for precise monitoring and control.
- **Cost-Effective Leg Massager:** Designed a leg massager using a cuff inflation and deflation mechanism controlled by a relay and Arduino.
- **Microprocessor Project Using 8086:** Designed a circuit in which an ID input produces output according to saved data.
- **Tremor Fixer Gloves:** Designed gloves to help people with hand tremors perform daily tasks more easily.
- **Biometric Identification of Unknown Patient:** Enabled doctors to access essential information of an unconscious patient through biometric identification.

COURSES, CERTIFICATIONS AND AWARDS

- 1st Prize, Blind Check Competition organized by ASME at the university.
- MATLAB Onramp, Simulink Onramp, Circuit Simulation Onramp and Simscape Onramp (MathWorks).
- International training on Optical Coherence Tomography (RS-1) and Yellow Laser (YLC-500).
- Attended two ophthalmic conferences in Karachi.

SKILLS

- **Technical:** MATLAB, Python, Ansys, Autodesk Fusion 360 (CAD), Minitab, Multisim, Proteus, MS Office, reporting and documentation
- **Academic and professional:** Research, critical thinking, problem solving, training and presentation, management